



Deep separable convolutional network for remaining useful life prediction of machinery



Biao Wang, Yaguo Lei^{*}, Naipeng Li, Tao Yan

Key Laboratory of Education Ministry for Modern Design & Rotor-Bearing System, Xi'an Jiaotong University, Xi'an 710049, China

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ABSTRACT

Deep learning is gaining attention in data-driven remaining useful life (RUL) prediction of machinery because of its powerful representation learning ability. With the help of deep learning techniques, the machine degradation information can be mined more sufficiently and some promising RUL prediction results have been achieved in several studies recently. These deep learning-based prognostics approaches, however, have the following weaknesses: 1) Their prediction performance largely depends on the hand-crafted feature design. 2) The correlations of different sensor data are not explicitly considered in representation learning. To overcome the above weaknesses, a new deep prognostics network named deep separable convolutional network (DSCN) is proposed in this paper for RUL prediction of machinery. In the proposed DSCN, the monitoring data acquired by different sensors are directly used as the inputs of the prognostics network. Then, a separable convolutional building block with a residual connection is built based on separable convolutions and squeeze and excitation operations. Through stacking multiple separable convolutional building blocks, the high-level representations are automatically learned from the input data. Finally, the RUL is estimated by feeding the learned representations into the fully-connected output layer. The proposed DSCN is validated using the vibration data from accelerated degradation tests of rolling element bearings and the public degradation simulation data of turbine engines, respectively. The experimental results show that the proposed DSCN is able to provide accurate RUL prediction results based on the raw multi-sensor data and is superior to some existing data-driven prognostics approaches.

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1. Introduction

With the advancement of sensor technologies, telecommunications and computer science, industrial internet of things has been widely applied into health monitoring of machinery in recent years, and a considerable number of monitoring data are acquired by multiple sensors during machine operation. Such explosions of data bring new opportunities and huge challenges to remaining useful life (RUL) prediction of machines. Meanwhile, because of the complexity of modern mechanical systems, it is very difficult to establish a precise mathematical or physical prognostics model based on first principles or failure mechanisms [1,2]. Therefore, how to sufficiently mine valuable machine degradation information from massive monitoring data and accurately build the corresponding RUL prediction model have become an urgent research subject currently. Since data-driven RUL prediction is able to automatically infer causalities hidden in data and directly model the degradation

^{*} Corresponding author.

E-mail address: yaguolei@mail.xjtu.edu.cn (Y. Lei).

characteristics of a complex mechanical system, it may be a powerful solution to process the huge amount of monitoring data and provide accurate RUL prediction results [3,4].

In general, data-driven RUL prediction consists of four major processes: data acquisition, feature extraction and selection, degradation behavior learning and RUL estimation [1,5]. First of all, different kinds of monitoring data, e.g., vibration and acoustic emission signals, are acquired by various sensors installed on a monitored machine. Then, to reflect the health conditions of the machine, representative features are extracted from the acquired data using signal processing techniques [6,7]. However, some of these hand-crafted features may be insensitive to degradation processes or even cannot offer any useful information for RUL prediction. Therefore, feature selection is implemented to find sensitive ones from all of the hand-crafted features. After that, the selected features are input into machine learning models, e.g., Gaussian process regression (GPR) [8], support vector machine (SVM) [9] and artificial neural networks (ANNs) [10], to learn the degradation behavior of the machine. Finally, the RUL is estimated using these learned models. For instance, Loutas et al. [9] used Spearman correlation coefficient to select the most monotonic features, and these features were fed into ε -SVM to predict the RUL of rolling element bearings. Javed et al. [11] developed a prognostics approach for cutting tools and bearings, in which features extracted from vibration signals were selected based on monotonicity and trendability, and extreme learning machine was used to perform the RUL estimation tasks. Lim et al. [12] presented a time series histogram of features method to extract features exhibiting local degradation trends from multi-sensor data, and used multi-layer perceptron to estimate the RUL of turbofan engines.

Recently, deep learning [13] is gaining more and more attention in data-driven RUL prediction. Deep learning is a subset of machine learning that uses ANNs with multiple layers to deliver state-of-the-art accuracy in various classification or regression tasks. Different from traditional machine learning techniques, deep learning techniques, such as deep belief networks (DBNs) [14], convolutional neural networks (CNNs) [15] and long short-term memory (LSTM) networks [16], are able to automatically learn multiple levels of representations from raw input data, without introducing hand-coded rules or domain knowledge. Because of this powerful representation learning ability, deep learning has achieved great successes in many fields including automatic transmission [17], speech recognition [18], natural language understanding [19], drug discovery [20] and others [21,22]. For data-driven RUL prediction, several deep learning-based studies have been carried out. Ren et al. [23] employed autoencoder to fuse thirty-six time-domain features, and the fused features were input into a deep neural network to estimate the RUL of rolling element bearings. Deutsch et al. [24] extracted six time- and frequency-domain features from vibration signals and used DBN to predict the RUL of spiral bevel gears. Zhu et al. [25] combined wavelet transform and CNN to predict the bearing RUL, where wavelet transform was first used to extract the time-frequency features and then multi-scale CNN was employed to estimate the RUL. Zhao et al. [26] utilized CNN to extract local robust features from the acquired multi-sensor data and employed bidirectional LSTM to predict the wear depth of cutting tools.

Although deep learning has achieved promising results in RUL prediction of machinery, existing prognostics approaches suffer from the following limitations. 1) Their prediction performance largely depends on the hand-crafted feature design in RUL prediction of many machines. Correspondingly, it is necessary to take a lot of effort on feature extraction and selection for obtaining sensitive features. Such procedures, however, require explicit prior knowledge and extensive human labor. In particular, features used in prognostics are usually designed case by case and so they cannot be applied generally in different industrial cases. 2) Current deep prognostics models do not explicitly consider the correlations of different sensor data in representation learning. The data from different sensors contain different degrees of degradation information and reflect the fault interaction among different components. For instance, the faults of one component may induce faults of the others due to the frequent connect, and further the new faults will make the original faults severer. Correspondingly, this propagation and interaction of faults can be captured by the sensors installed on the components. Therefore, for accurate RUL prediction, it is important to capture the dependencies of different sensor data and emphasize the information that is most relevant to the health conditions of machines.

To deal with the above-mentioned limitations, a new deep prognostics network, named deep separable convolutional network (DSCN), is proposed for RUL prediction of machinery in this paper. In the proposed DSCN, the monitoring data acquired by different sensors are directly used as the inputs of the prognostics network. Then, a separable convolutional building block is built based on separable convolutional layers and squeeze and excitation (SE) unit. Through stacking multiple separable convolutional building blocks, the high-level representations can be automatically learned from the raw input data. Finally, the RUL is estimated by feeding the learned representations into the last fully-connected layer. Two case studies, i.e., RUL prediction of rolling element bearings and RUL prediction of turbofan engines, are carried out to verify the proposed DSCN. Experimental results show that the proposed DSCN achieves high RUL prediction accuracy and outperforms some traditional data-driven approaches and typical deep learning models. The main contributions of this paper are summarized as follows.

- 1) A separable convolution operation is introduced into RUL prediction of machinery. The introduction of separable convolutions not only effectively models the interrelationships of different sensor data, but also greatly reduces the size of the prognostics network.
- 2) An SE unit is constructed to adaptively recalibrate the feature responses from the separable convolutional layer. The dynamic feature response recalibrations highlight informative feature maps and suppress useless ones, thus enhancing the information discriminability of the prognostics network.

- 3) A new deep prognostics network, i.e., DSCN, is proposed by building and stacking multiple separable convolutional building blocks based on the separable convolutions and the SE units. The proposed DSCN is able to provide accurate RUL estimations based on raw multi-sensor data of machinery and is superior to some existing prognostics approaches.

The rest of this paper is organized as follows. Section 2 briefly introduces standard convolutional networks and residual connections. Section 3 details the proposed DSCN for RUL prediction of machinery. In Section 4, the effectiveness and superiority of the proposed DSCN are validated using vibration data from accelerated degradation tests of rolling element bearings and public Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) datasets, respectively. Finally, conclusions are drawn in Section 5.

2. Preliminaries

In this section, a basic architecture of convolutional networks is first briefly introduced, and then residual connections for convolutional networks are reviewed.

2.1. Convolutional networks

Convolutional networks, also known as CNNs or ConvNets, are one particular type of deep feed-forward networks that is designed to process multiple array data, such as time-series signals, images and audio spectrograms. They are structured by a stack of a few distinct types of layers, including convolutional layer, pooling layer and fully-connected layer [15].

1) *Convolutional layer*. The convolutional layer is the core building block of convolutional networks. A convolutional layer is composed of different feature maps that are produced by convolution operations and activation operations. Each neuron of a feature map is only connected to a local region of the input volume through a set of weights called a kernel. All neurons in a feature map share the same kernel, thus reducing model parameters and making the network more efficient to implement. Mathematically, the n -th feature map of the l -th convolutional layer, $\mathbf{x}_n^l$, can be calculated by

$$\mathbf{x}_n^l = \sigma(\mathbf{z}_n^l) \quad (1)$$

$$\mathbf{z}_n^l = \mathbf{k}_n^l * \mathbf{x}^{l-1} + \mathbf{b}_n^l = \sum_{c=1}^C \mathbf{k}_{n,c}^l * \mathbf{x}_c^{l-1} + \mathbf{b}_n^l \quad (2)$$

where $\mathbf{z}_n^l$ is the output of convolution operations, $\sigma(\cdot)$ is the nonlinear activation function, such as sigmoid, tanh and rectified linear unit (ReLU) [13], $*$ denotes the convolution operator, $\mathbf{k}_n^l$ is the n -th convolution kernel, i.e., the n -th weight vector, $\mathbf{b}_n^l$ is the bias term, $\mathbf{x}^{l-1}$ is the input volume, i.e., the feature maps in the previous layer $l-1$, and C is the number of input channels.

2) *Pooling layer*. The pooling layer aims to merge semantically similar features into one by using nonlinear down-sampling functions like max pooling, average pooling, and L^2 -norm pooling. The pooling operation is independently conducted on every input feature map and outputs a summary statistic of a local patch of the previous feature map, thereby decreasing the spatial size of representations. In the l -th pooling layer, the n -th feature map $\mathbf{x}_n^{l-1}$ can be obtained by

$$\mathbf{x}_n^l = \text{pool}(\mathbf{x}_n^{l-1}, p, s) \quad (3)$$

where $\mathbf{x}_n^{l-1}$ is the n -th input feature map of the l -th pooling layer, i.e., the n -th feature map in the previous layer $l-1$, $\text{pool}(\cdot)$ is the down-sampling function, p is the pooling size, and s is the stride size.

3) *Fully-connected layer*. The fully-connected layer is the same as the traditional multi-layer perceptron in principle, where each neuron is fully connected to all neurons in the previous layer. In general, the fully-connected layer is used as the output layer of convolutional networks to perform the high-level reasoning. For the l -th fully-connected layer, its output $\mathbf{x}^l$ can be computed by

$$\mathbf{x}^l = \sigma(\mathbf{w}^l \mathbf{x}^{l-1} + \mathbf{b}^l) \quad (4)$$

where $\mathbf{x}^{l-1}$ is the input of the l -th fully-connected layer, i.e., the output of the previous layer $l-1$, $\mathbf{w}^l$ is the weight matrix, and $\mathbf{b}^l$ is the bias vector.

2.2. Residual connections for convolutional networks

Residual connections are first proposed by He et al. in [27], aiming at reducing the training complexity of convolutional networks and enabling them to be substantially deeper. The network depth, i.e., the number of stacked layers plays a crucial role in enhancing the representation ability of convolutional networks. However, with the increase of the network depth, the training is more difficult and the accuracy gets saturated or even decreases rapidly. To address this problem, residual connections are introduced to the convolutional networks in [27]. In a convolutional building block, the residual connection is

established by attaching an identity shortcut between the input and the output. Correspondingly, instead of learning an unreferenced mapping, the building block with a residual connection learns a residual mapping with respect to the input, which is easier to be optimized than the original mapping. Meanwhile, residual connections add neither extra parameters nor computational complexity. More importantly, the convolutional networks using residual connections are able to easily achieve performance improvement by increasing the network depth. Formally, let $\mathcal{H}(\mathbf{x}^{l-1}, \mathbf{W}^l)$ denote the desired underlying mapping of the building block with a residual connection, and then the stacked weight layers in this building block will fit a residual mapping of $\mathcal{F}(\mathbf{x}^{l-1}, \mathbf{W}^l) := \mathcal{H}(\mathbf{x}^{l-1}, \mathbf{W}^l) - \mathbf{x}^{l-1}$. Accordingly, the output of the building block can be expressed as

$$\mathbf{x}^l = \mathbf{x}^{l-1} + \mathcal{F}(\mathbf{x}^{l-1}, \mathbf{W}^l) \quad (5)$$

where $\mathbf{x}^{l-1}$ and $\mathbf{x}^l$ are respectively the input and output of the building block, $\mathcal{F}(\cdot)$ is the residual function, e.g., a stack of two convolutional layers, $\mathbf{W}^l = (\mathbf{W}_1^l, \mathbf{W}_2^l, \dots, \mathbf{W}_L^l)$ is the learnable layer parameters, i.e., weights and biases, and L is the number of the stacked layers in the building block.

3. Proposed DSCN for RUL prediction of machinery

This section details the proposed DSCN for RUL prediction of machinery. The proposed DSCN directly uses the monitoring data from different sensors as inputs. Then, to effectively model interrelationships of different sensor data, separable convolutions are introduced to replace the standard convolutions. Meanwhile, to increase the sensitivity of DSCN to informative feature maps, a new architecture unit, called SE unit, is constructed behind the separable convolutional layer to perform adaptive feature response recalibrations. After that, the core building block of DSCN, separable convolutional building block, is built using the separable convolutions and the SE unit. By stacking multiple separable convolutional building blocks, the high-level representations are learned from the input data, and the RUL is finally estimated by feeding the learned representations into a fully-connected layer.

3.1. Separable convolutions

In order to integrate the degradation information from different sensors, multi-channel time-series data are usually used as the inputs of deep prognostics models, where each channel represents one sensor sequence. For these multi-sensor time-series input data, the data in each channel have temporal correlations, because they are acquired by the same sensor and reflect the failure progression of one monitored component over operating time. Meanwhile, the data in different channels also have cross-channel correlations because of the propagation and interaction of faults among different components. Existing deep prognostics models, however, synchronously map temporal correlations and cross-channel correlations to model the degradation processes of machinery [26,28–30], and thus are unable to effectively capture and model the dependencies of different sensor data, which limits the prediction accuracy of the prognostics models and their generalization. Therefore, instead of the standard convolution operation described in Eq. (2), a separable convolution operation [31] is introduced into the RUL prediction of machinery in this paper, aiming to effectively model interrelationships of different sensor data by decoupling temporal correlations and cross-channel correlations.

The separable convolutions unravel temporal correlations and cross-channel correlations by factorizing the standard convolutions. As shown in Fig. 1, the standard convolutions are decomposed into two parts: channel-wise convolutions and pointwise convolutions. At first, Channel-wise convolutions apply a single convolution kernel into each of the input channels to separately map temporal correlations of every sensor sequence. Then, pointwise convolutions perform 1×1 convolutions to create a linear combination of the outputs of the channel-wise convolutions, i.e., to map cross-channel correlations of different sensors. Through the above two independent steps, the temporal correlations and cross-channel correlations can be sufficiently decoupled. In particular, separable convolutions use a specific convolution kernel, i.e., pointwise convolution kernel, to map the cross-channel correlations, which effectively captures the dependencies of different sensor data.

Let $\mathbf{x}^{l-1} \in \mathbb{R}^{H \times W \times C}$ denote the input volume, where H and W are the height and width of the input volume respectively, and C is the number of input channels. Especially, for multi-sensor input data, e.g., vibration signals from different accelerators, H is the length of each sensor sequence, W is equal to 1, and C is the number of sensors. Correspondingly, the convolution kernels in a standard convolutional layer are denoted as $\mathbf{K} \in \mathbb{R}^{M \times 1 \times C \times N}$, where $M \times 1$ is the size of a convolution kernel, and N is the number of kernels, i.e., the number of output channels. In the separable convolutions, $\mathbf{K}$ is factorized into the channel-wise convolution kernels $\mathbf{R} \in \mathbb{R}^{M \times 1 \times C}$ and the pointwise convolution kernels $\mathbf{P} \in \mathbb{R}^{C \times N}$. Therefore, the n -th output of separable convolutions, $\mathbf{z}_n^l$, can be given by

$$\mathbf{y}_c^l = \mathbf{R}_c * \mathbf{x}_c^{l-1} + \mathbf{b}_c^l \quad (6)$$

$$\mathbf{z}_n^l = \sum_{c=1}^C \mathbf{P}_n * \mathbf{y}_c^l + \mathbf{b}_n^l \quad (7)$$

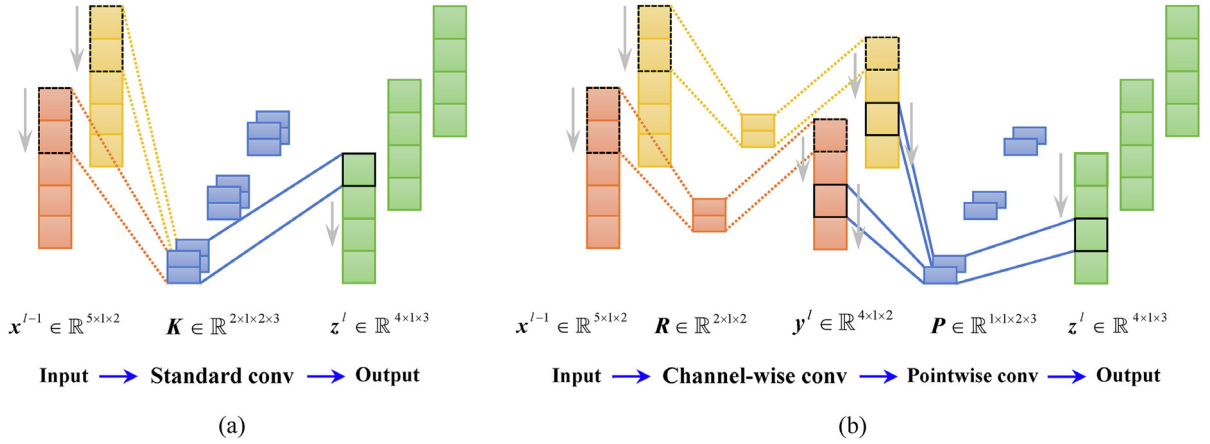


Fig. 1. An illustration for convolution operations. (a) Standard convolutions. (b) Separable convolutions.

where y_c^l is the c -th output of the channel-wise convolutions, R_c is the c -th channel-wise convolution kernel that is applied to the c -th channel of x^{l-1} , b_c^l is the bias vector corresponding to R_c , P_n is the n -th pointwise convolution kernel of size 1×1 , and b_n^l is the bias vector corresponding to P_n .

3.2. Feature response recalibrations

The feature maps in a separable convolutional layer contain different degrees of representative information. Accordingly, some of these feature maps may be informative, but others may be not. To highlight informative feature maps and suppress useless ones, an SE unit is constructed behind the separable convolutional layer in this paper. As shown in Fig. 2, the SE unit is able to adaptively recalibrate the responses of each feature map by evaluating its informativeness. To be specific, the process of feature response recalibrations is comprised of two steps, including squeeze operation and excitation operation. Firstly, the squeeze operation embeds the global information from all output channels of the separable convolutional layer into a channel descriptor $u^l \in \mathbb{R}^C$. This step is achieved by using global average pooling to shrink the convolutional output z^l , where u^l consists of N channel-wise statistics, and the n -th element of u^l can be calculated by

$$u_n^l = \frac{1}{H} \sum_{h=1}^H z_{n,h}^l \quad (8)$$

Then, the excitation operation utilizes a self-gating mechanism [32] to estimate the informativeness of every channel based on the channel descriptor u^l , generating the corresponding channel weights ω^l , i.e.,

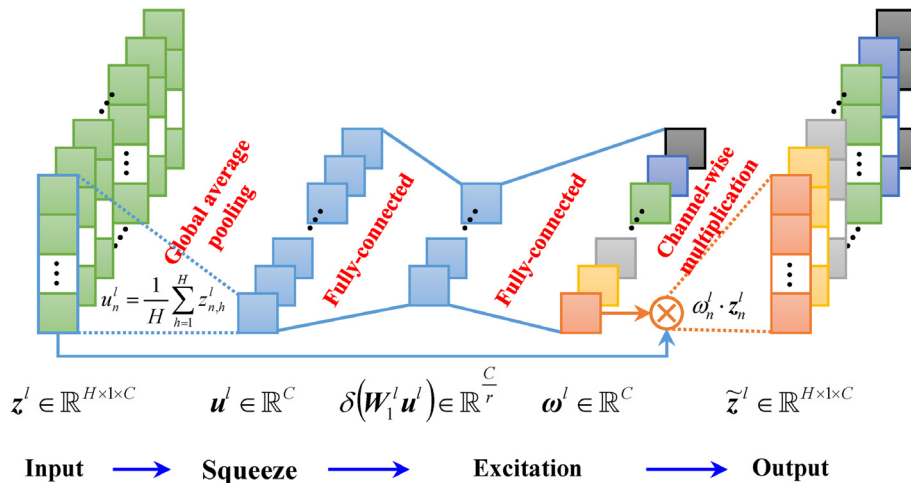


Fig. 2. An illustration for feature response recalibrations in an SE unit.

$$\omega^l = \sigma\left(\mathbf{W}_2^l \delta\left(\mathbf{W}_1^l \mathbf{u}^l\right)\right) \quad (9)$$

where $\sigma(\cdot)$ and $\delta(\cdot)$ are sigmoid and ReLU activation functions respectively, $\mathbf{W}_1^l \in \mathbb{R}^{\frac{C}{r} \times C}$ and $\mathbf{W}_2^l \in \mathbb{R}^{C \times \frac{C}{r}}$ are the weights, and r is the ratio of dimensionality reduction. To limit the computational complexity, the self-gating mechanism is parameterized by two fully-connected layers, including a dimensionality reduction layer with $\frac{C}{r}$ neurons and a dimensionality increase layer with C neurons. Finally, the recalibrated feature maps are obtained by conducting channel-wise multiplication between the feature map $\mathbf{z}_n^l$ from the separable convolutional and the corresponding channel weight ω_n^l .

3.3. Separable convolutional building blocks

As discussed in Section 2.2, the representation ability of convolutional networks can be enhanced by stacking more weight layers. However, the increase of the network depth makes the training more difficult, resulting in accuracy saturation or even accuracy reduction [27]. Thus, this paper employs residual connections to ease the training of DSCN. Meanwhile, the pre-activation strategy in [33] is also used to improve the regularization of DSCN so as to relieve the problem of overfitting. Fig. 3 displays the separable convolutional building block with a residual connection in the proposed DSCN, where an identity skip connection is added between the input $\mathbf{x}^{l-1}$ and the output $\mathbf{x}^l$. The separable convolutional building block consists of two separable convolutional layers, an average pooling layer and an SE unit. In particular, both batch normalization (BN) [34] and ReLU [35] are used as pre-activation and are added before each separable convolution.

In the separable convolutional building block, the use of BN is to make the input distribution of each separable convolution more stable and reduce the overfitting [34]. Further, BN is able to allow DSCN to use much higher learning rates and to be insensitive to the parameter initialization. For the input $\mathbf{x}^{l-1} = (\mathbf{x}_1^{l-1}, \mathbf{x}_2^{l-1}, \dots, \mathbf{x}_C^{l-1})$, BN performs the normalization for each input channel $\mathbf{x}_c^{l-1}$, and the transformation of BN is able to be written as

$$\hat{\mathbf{x}}_c^{l-1} = \frac{\mathbf{x}_c^{l-1} - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}} \quad (10)$$

$$\mathbf{y}_c^{l-1} = \gamma_c \hat{\mathbf{x}}_c^{l-1} + \beta_c \quad (11)$$

where $\mathbf{y}_c^{l-1}$ is the transformation output corresponding to $\mathbf{x}_c^{l-1}$, μ_B and σ_B^2 are respectively the expectation and variance of a mini-batch B in the c -th input channel, ε is a constant added to σ_B^2 for numerical stability, and γ_c is the c -th learnable scale parameter in the BN transformation, and β_c is the c -th learnable shift parameter.

In general, BN and ReLU are added after convolution operations in the standard convolutional networks, but this post-activation strategy may not take full advantage of the benefits of BN in the convolutional networks with residual connections [33]. Hence, the full pre-activation strategy is adopted in this paper, i.e., BN and ReLU are conducted before each separable convolution. Moreover, to reduce the dimension of the representation, an average pooling layer is inserted before the SE unit to perform down-sampling. Subsequently, the feature maps recalibrated by the SE unit are added to the identity mapping $\mathbf{x}^{l-1}$ to generate the output of the separable convolutional building block $\mathbf{x}^l$, which is then fed into next separable convolutional building blocks or subsequent processing layers.

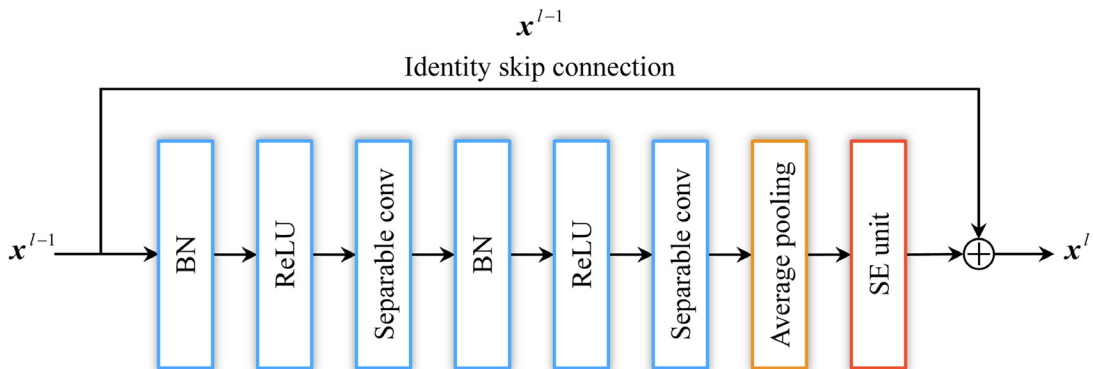


Fig. 3. A separable convolutional building block with a residual connection in the proposed DSCN.

3.4. Architecture of the proposed DSCN

Fig. 4 illustrates the architecture of the proposed DSCN, in which the input is monitoring data acquired by different sensors and the output is the corresponding RUL value. DSCN is comprised of two sub-networks, i.e., representation learning sub-network and RUL estimation sub-network. The representation learning sub-network is able to automatically discover the discriminative information and learn multiple levels of representations from the input data by stacking multiple separable convolutional building blocks, and the learned high-level representations are fed into the RUL estimation sub-network to perform RUL prediction through a fully-connected layer. More details of DSCN are given as follows.

1) The representation learning sub-network firstly uses a separable convolutional layer to convolute the input sensor data of size $H \times 1 \times C$, in which the size of a channel-wise convolution kernel is set to $M \times 1$, the number of pointwise convolution kernels, i.e., the number of output channels, is set to N , and both BN and ReLU are adopted right after separable convolutions. Following the separable convolutional layer, an average pooling layer is employed to conduct down-sampling, where the pooling size p is equal to the stride size s . Afterwards, the learned low-level representations are forwarded to succeeding separable convolutional building blocks to obtain higher-level representations. For the d -th separable convolutional building block, where $d = 1, 2, \dots, D$, the configuration of two separable convolutional layers is the same, i.e., each channel-wise convolution kernel has size $M \times 1$, and a total of $2^d N$ pointwise convolution kernels are contained in each separable convolutional layer. Additionally, the average pooling is performed using a non-overlapping window, i.e., $p = s$, and the dimensionality reduction ratio r in the SE unit is set to 16 by considering the trade-off between prediction accuracy and computational complexity, which will be discussed in Section 4.1.5.

2) The RUL estimation sub-network utilizes a global average pooling layer to receive the high-level representations from the representation learning sub-network, i.e., the output of the D -th separable convolutional building block. Correspondingly, $2^D N$ input feature maps from the D -th separable convolutional building block are transformed into a vector of size $2^D N$. It is to be noted that the use of global average pooling is to reduce the total number of parameters. After that, a fully-connected layer is used as the output layer of DSCN to estimate the RUL, in which only one neuron is attached at the end of the proposed network and ReLU is adopted to implement the nonlinear activation.

4. Experimental verification

To verify the effectiveness and superiority of the proposed DSCN in coping with the degradation problems of mechanical components or systems, two case studies are carried out in this section, including: (1) RUL prediction of rolling element bearings; and (2) RUL prediction of turbofan engines. In the first case study, vibration signals collected from accelerated degradation tests of bearings are used to evaluate the prediction performance of DSCN, and four state-of-the-art prognostics approaches are employed for comparison purposes as well. In the second case study, the proposed DSCN is further validated using the prognostics benchmark datasets, i.e., the public C-MAPSS datasets, and is also compared with six published studies that use the same validation datasets.

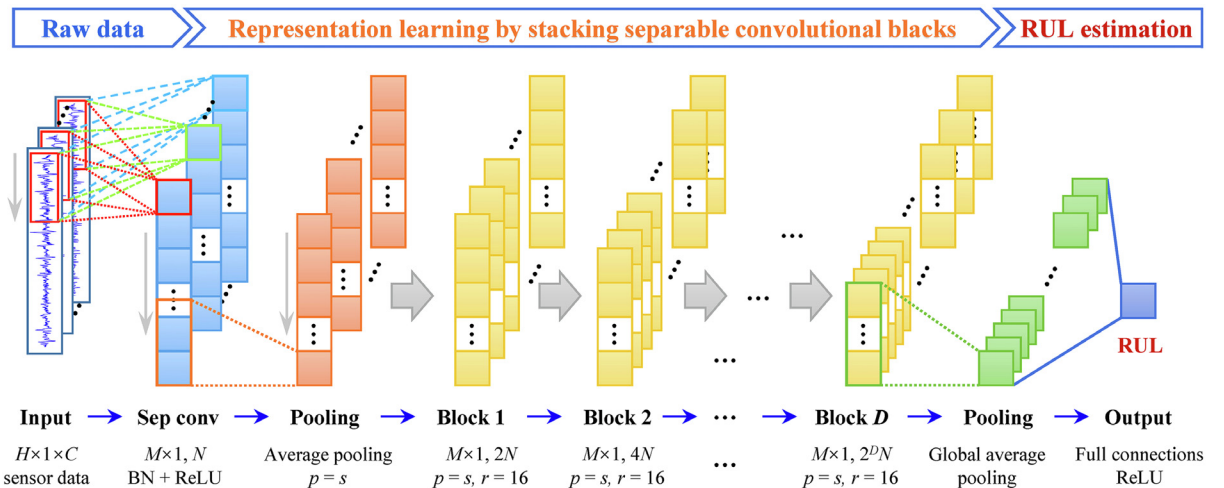


Fig. 4. Architecture of the proposed DSCN.

4.1. Case study 1: Accelerated degradation experiments of rolling element bearings

4.1.1. Data description

Fig. 5 displays the testbed of rolling element bearings [36], which is composed of an alternating current (AC) motor, a motor speed controller, a support shaft, two support bearings (heavy duty roller bearings), a hydraulic loading system and so on. This experimental platform is able to conduct accelerated degradation tests of bearings under different operating conditions and acquire complete run-to-failure data. In this paper, as tabulated in Table 1, a total of fifteen LDK UER204 ball bearings are tested under three different operating conditions. The radial force is generated by the hydraulic loading system and applied to the housing of tested bearings, and the rotating speed is set and kept by the speed controller of AC induction motor. Fig. 6 shows the photographs of normal and degraded bearings. It can be observed that the failure of the tested bearings is caused by different types of faults, including inner race wear, cage fracture, outer race wear, outer race fracture, etc. To collect vibration signals of the tested bearings, as shown in Fig. 5, two accelerometers of type PCB 352C33 are positioned at 90° on the housing of the tested bearings, i.e., one is placed on the horizontal axis and the other one is placed on the vertical axis. The sampling frequency is set to 25.6 kHz, and 32,768 data points (i.e. 1.28 s) are recorded each 1 min. Fig. 7 plots the horizontal and vertical vibration signals under three different operating conditions. In this paper, the first four bearing data-sets of each operating condition are selected as training datasets, and the remaining one is testing dataset. In particular, both horizontal and vertical vibration signals are used as the inputs of DSCN.

4.1.2. Evaluation metrics

In this paper, the prediction performance of DSCN is quantitatively evaluated by employing scoring function and root mean square error (RMSE), which are introduced as follows.

1) *Scoring function*: The scoring function used in this paper comes from the 2008 Prognostics and Health Management Data Challenge [37], which is defined as

$$\text{score} = \begin{cases} \sum_{i=1}^Q (e^{\frac{d_i}{13}} - 1) & \text{for } d_i < 0 \\ \sum_{i=1}^Q (e^{\frac{d_i}{10}} - 1) & \text{for } d_i \geq 0 \end{cases} \quad (12)$$

where *score* is the score value of RUL prediction results, *Q* is the number of testing samples, and $d_i = \text{PreRUL}_i - \text{ActRUL}_i$ is the error between the estimated RUL and the actual RUL with respect to the *i*-th testing sample. Fig. 8 depicts the scoring function, in which the error value ranges from −50 to 50 with an interval of 2. From Fig. 8, it can be seen that the smaller the absolute error value is, the lower the score is. Furthermore, the scoring function is asymmetric so that late predictions (i.e., $d_i \geq 0$) are more heavily penalized than early predictions (i.e., $d_i < 0$). However, it should be noted that since the penalty

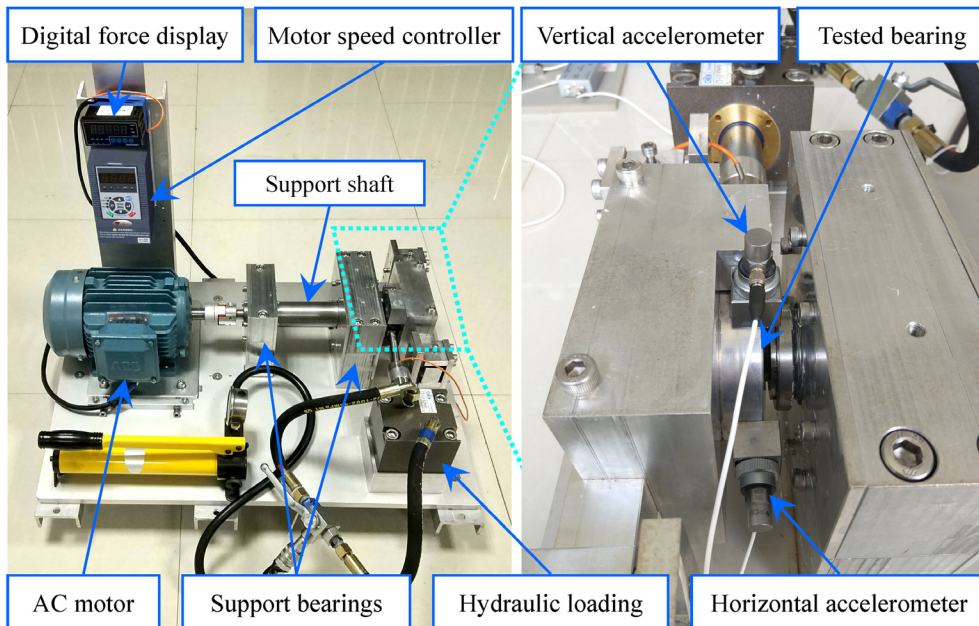


Fig. 5. Testbed of rolling element bearings.

Table 1
Operating conditions of tested bearings.

Operating condition	Radial force	Rotating speed	Bearing dataset	
			Training dataset	Testing dataset
Condition 1	12 kN	2100 rpm	Bearing 1_1 Bearing 1_2 Bearing 1_3 Bearing 1_4	Bearing 1_5
Condition 2	11 kN	2250 rpm	Bearing 2_1 Bearing 2_2 Bearing 2_3 Bearing 2_4	Bearing 2_5
Condition 3	10 kN	2400 rpm	Bearing 3_1 Bearing 3_2 Bearing 3_3 Bearing 3_4	Bearing 3_5

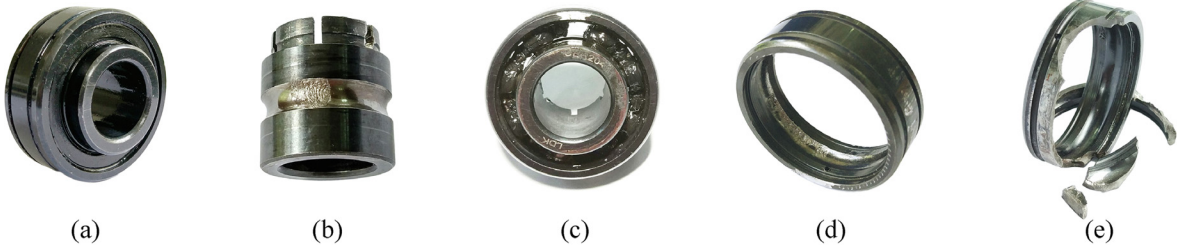


Fig. 6. Photographs of normal and degraded bearings. (a) Normal bearing. (b) Inner race wear. (c) Cage fracture. (d) Outer race wear. (e) Outer race fracture.

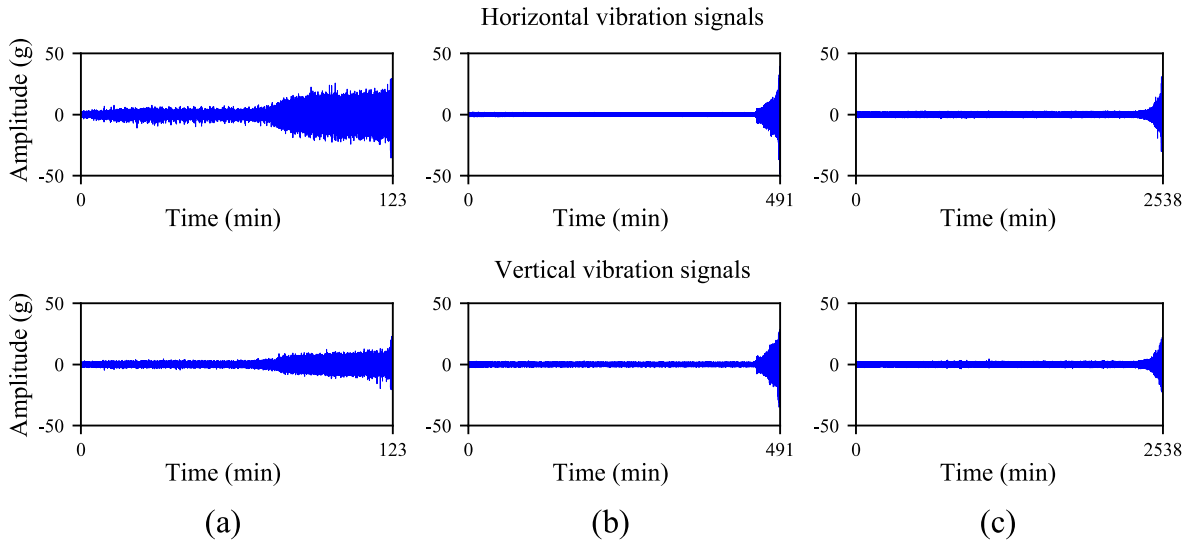


Fig. 7. Horizontal and vertical vibration signals. (a) Bearing 1_1. (b) Bearing 2_1. (c) Bearing 3_1.

grows exponentially with the increased error, the score may be entirely dominated by a single outlier with a very large prediction error. To address this problem, RMSE is introduced as the second performance evaluation metric in this paper.

2) **RMSE**: RMSE is a commonly used evaluation metric in the field of prognostics and health management. Given RUL prediction results, RMSE value is calculated by

$$RMSE = \sqrt{\frac{1}{Q} \sum_{i=1}^Q d_i^2} \quad (13)$$

Fig. 8 also shows RMSE as a function of the error. It can be observed that, different from the scoring function, RMSE assigns an equal weight to each prediction, thereby avoiding the interference of outliers.

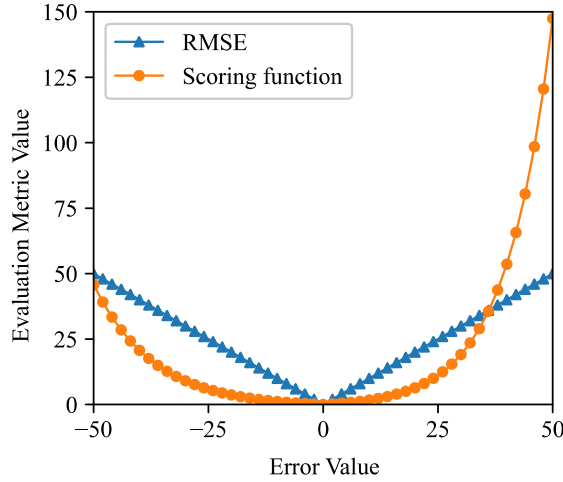


Fig. 8. Comparison between scoring function and RMSE using different error values.

4.1.3. Data preprocessing

1) *Normalization*: For horizontal and vertical vibration signals, the data acquired at each sampling time step are normalized using z-score, i.e.,

$$\mathbf{x}_{\text{norm}}^t = \frac{\mathbf{x}^t - \mu^t}{\sigma^t} \quad (14)$$

where $\mathbf{x}^t$ represents the original vibration signals collected at t -th sampling time step, $\mathbf{x}_{\text{norm}}^t$ is the normalized vibration signals corresponding to $\mathbf{x}^t$, and μ^t and σ^t denote the mean and standard deviation of $\mathbf{x}^t$, respectively. In addition, it is to be noted that the tested bearings have a very wide range of lifetime (from 33 min to 2538 min). If the actual RUL value at each sampling time step is used as the desired output of DSCN during training, the significant difference in the life duration will lead to the problem of underfitting [38]. Therefore, during the data pre-processing, it is also an essential step to normalize the actual RUL values of each bearing in the training datasets to be within the range of [0, 1] via dividing by their respective bearing lifetime. Then, these normalized RUL values are used as the target outputs of DSCN. Correspondingly, based on the definition of RUL [39], the estimated RUL values of each bearing in the testing datasets can be obtained by

$$\text{PreRUL}_t = \frac{\text{NormRUL}_t}{1 - \text{NormRUL}_t} \times P_t \quad (15)$$

where P_t is the inspection time corresponding to the t -th sampling time step, NormRUL_t is the predictive value of the trained DSCN at P_t , and PreRUL_t is the estimated RUL at P_t .

2) *Time window embedding*: For data-driven prognostics problems, how to embed useful temporal information into the inputs of prediction models is an important consideration. If the prediction models only use the data acquired at a single sampling time step as an input, the preceding temporal information that is related to current degradation state will be neglected, therefore restricting the prediction performance of the model. To deal with such an issue, this paper adopts a time window embedding strategy [40] to process the vibration signals of the tested bearings after normalization, in which a fixed-size time window is utilized to concatenate the vibration signals acquired at consecutive sampling time steps into a high-dimensional vector and then this vector is fed as an input to DSCN. Accordingly, at each sampling time step, the input vector obtained by time window embedding consists of the vibration signals sampled at current time step and its previous $S - 1$ time steps, which can be formulated as

$$\mathbf{x}_{\text{input}}^t = (\mathbf{x}_{\text{norm}}^{t-S+1}, \dots, \mathbf{x}_{\text{norm}}^{t-1}, \mathbf{x}_{\text{norm}}^t) \quad (16)$$

where S is the size of the time window. Fig. 9 shows an example of time window embedding, in which a time window of size 3 is used to enclose the vibration signals into an input vector at each time step.

4.1.4. Configuration of DSCN

In the proposed DSCN, the hyperparameters which need to be predefined contain the channel-wise convolution kernel size $M \times 1$, the number of pointwise convolution kernels N , the pooling size p , the dimensionality reduction ratio of SE units r , and the number of separable convolutional building blocks D . For RUL prediction of bearings, these hyperparameters are determined by performing 4-fold cross-validation on the training datasets, and Table 2 lists the final hyperparameter selection results, which is determined by considering the trade-off between the prediction accuracy and the computational cost. Especially, the following Section 4.1.5 will detailedly discuss the effects of the reduction ratio r and the number of blocks D

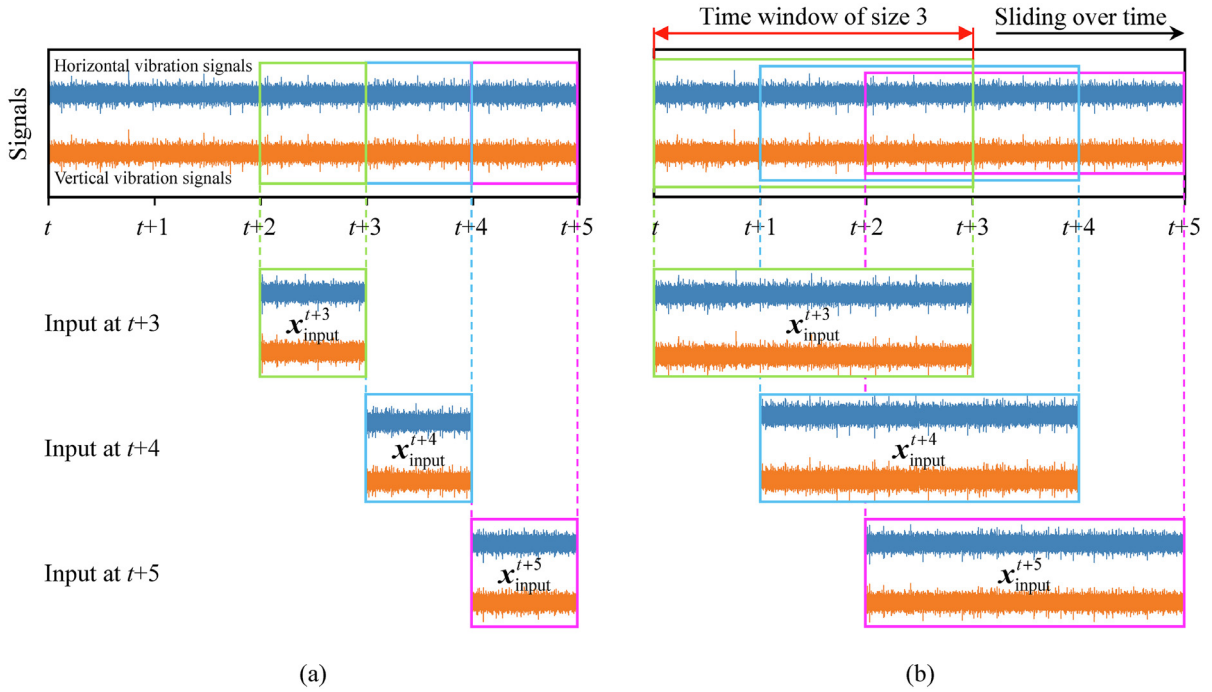


Fig. 9. Illustration of time window embedding. (a) No time window. (b) Time window of size 3.

Table 2

Configuration of DSCN in RUL prediction of bearings.

Hyperparameter	Size	Hyperparameter	Size
Kernel size $M \times 1$	8×1	Number of kernels N	16
Pooling size p	4	Reduction ratio r	16
Number of blocks D	3	Time window size S	5
Mini-batch size	128	Number of epochs	100

on the prediction performance of DSCN. Moreover, a time window of size 5 is employed to process the normalized vibration signals. Here, each normalized signal sequence $\mathbf{x}_{\text{norm}}^t$ contain 2560 data points and correspondingly the size of an input volume is $12800 \times 1 \times 2$. During training, this paper uses mean square error (MSE) as the loss function of DSCN, and Adam optimizer [41] with a mini-batch size of 128 is adopted to minimize the MSE loss function by iteratively updating the network weights and biases. The model is trained for 100 epochs from scratch, and the weights and biases are initialized using the strategy described in [27].

4.1.5. Experimental results

In this case study, some factors that may affect the prediction performance of DSCN are first investigated in detail, including the time window size, the dimensionality reduction ratio and the network depth. Then, the benefits from separable convolutions and feature response recalibrations are analyzed and discussed. Finally, the proposed DSCN is compared with four state-of-the-art prognostics approaches to demonstrate its superiority. Each experiment in the following studies is repeated 20 times to reduce the effect of randomness. In particular, for each bearing in the testing datasets, the score and RMSE are calculated from the half to the end of lifetime, because the predictions at these inspection time instances are more reliable and meaningful than the early stage [42].

1) *Analysis of time window embedding*: As discussed in the above section, the time window embedding is able to gather more useful degradation information into the inputs of DSCN and so improves its prediction performance. In particular, this performance improvement is related to the time window size. Therefore, this section first analyzes the influence of different time window sizes on the prediction performance of DSCN. For RUL prediction of bearings, the time window size S is set to 1, 3, 5, 7 and 9, respectively. Correspondingly, the box plots of the scoring function values and the RMSE values of three tested bearings are separately shown in Figs. 10 and 11. From these box plots, it is observed that the larger the time window size is, the lower the score and RMSE values are, which implies that the prediction performance of DSCN can be effectively enhanced by increasing the time window size to aggregate more time-series information about bearing degradation. Further, Fig. 12

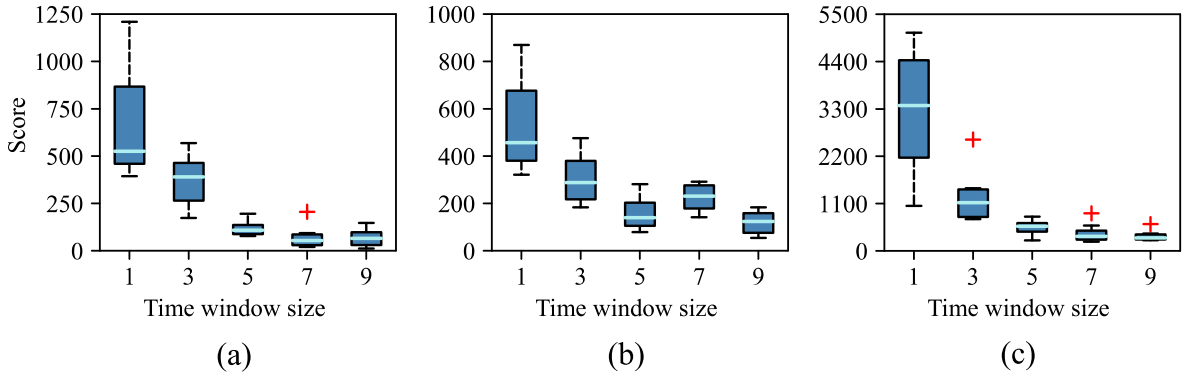


Fig. 10. Box plots of scoring function values under different time window sizes. (a) Bearing 1_5. (b) Bearing 2_5. (c) Bearing 3_5.

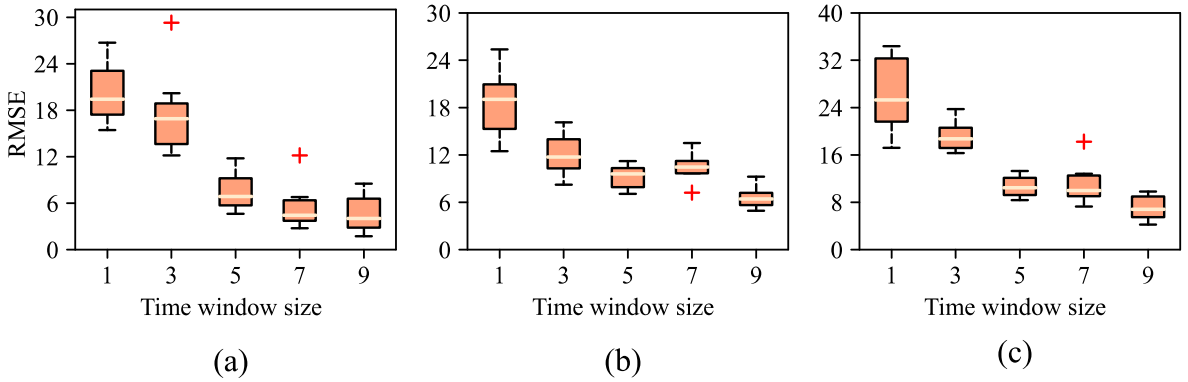


Fig. 11. Box plots of RMSE values under different time window sizes. (a) Bearing 1_5. (b) Bearing 2_5. (c) Bearing 3_5.

shows the RUL prediction results of Bearing 3_5 under six different time window sizes. It can be clearly seen that the increase of time window size makes RUL prediction more accurate while reducing the fluctuations of the estimated RUL values. However, it should be noted that as the time window size increases, the room for performance improvement gradually reduces, but more memory storage and computation time are required because a large time window size will generate high-dimensional input vectors. Based on the above analyses, the time window size S is set to 5 in RUL prediction of bearings.

2) *Effect of dimensionality reduction ratio and network depth*: The dimensionality reduction ratio r is a critical hyperparameter in the SE units, whose size may affect the capacity and computational cost of SE units. To investigate this influence, DSCN with different dimensionality reduction ratios is used to perform RUL estimation of bearings, and the final estimation results are reported in Table 3. It can be observed that although the increase of the dimensionality reduction ratio reduces the total model parameters and training time, the performance of DSCN cannot be improved monotonically. Particularly, when the dimensionality reduction ratio is equal to 16, the prediction model achieves a good trade-off between accuracy and computational cost. Accordingly, this paper set the dimensionality reduction ratio r to be 16. Furthermore, this section also investigates the effect of the network depth on the prediction performance of DSCN. For the proposed DSCN, the network depth can be increased by stacking more separable convolutional building blocks. Here, a total of five different network depths are studied and the corresponding performance estimation results are summarized in Table 4. From this table, it can be seen that deeper prognostics network gets lower score values and smaller RMSE values in RUL prediction of bearings. This means that the increment of the network depth boosts the representation learning ability of DSCN. However, the deeper architecture will cause a heavier computational burden. Meanwhile, the limited training samples may lead to overfitting as the prognostics network become deeper [28]. Consequently, the number of separable convolutional building blocks D is set to 3 in this paper.

3) *Benefits of separable convolutions and feature response recalibrations*: To illustrate the benefits from separable convolutions and feature response recalibrations described in Section 3, the other two prognostics networks are also used to perform RUL prediction of bearings in this section, including the network with standard convolutions (denoted as standard ConvNet) and the network with separable convolutions (denoted as separable ConvNet). Except for not containing the SE units, these two prognostics networks have the same architecture and hyperparameter settings as the proposed DSCN. Table 5 reports the total model parameters and training time of these three different prognostics networks. It can be observed that compared with standard convolutions, separable convolutions greatly reduce the model size (82.07% reduction), resulting in a

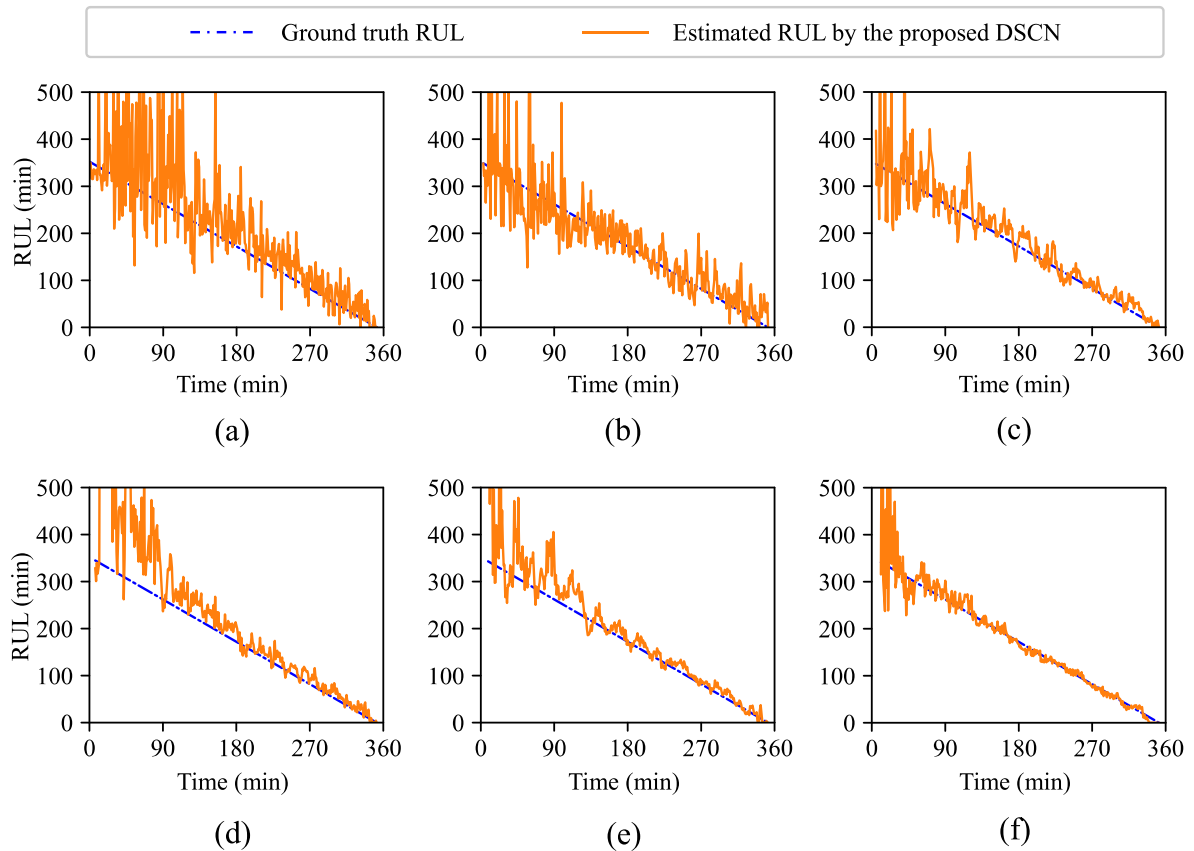


Fig. 12. RUL prediction results of Bearing 3_5 using different time window sizes. (a) $S = 1$. (b) $S = 3$. (c) $S = 5$. (d) $S = 7$. (e) $S = 9$. (f) $S = 11$.

lower computational cost. Whereas for the proposed DSCN, the introduction of SE units only increases 5.50% parameters in comparison to the separable ConvNet. Fig. 13 shows the performance estimation results of these three different prognostics networks. It can be seen that separable ConvNet obtains a lower score value and a smaller RMSE value than standard ConvNet for each bearing, which means that through sufficiently decoupling temporal correlations and cross-channel correlations, separable convolutions effectively improve the performance of the prognostics network. Further, by employing the SE units to adaptively recalibrate the feature responses from the separable convolutional layers, the proposed DSCN achieves better prediction performance than the other two prognostics networks. As a result, benefiting from separable convolutions and SE units, the proposed DSCN has a clear advantage in accuracy and computational complexity as compared to the standard convolutional networks. Next, the superiority of the proposed DSCN will be further demonstrated by comparing with four state-of-the-art prognostics approaches.

4) *Comparison with state-of-the-art prognostics approaches*: In this section, four existing prognostics approaches are implemented to predict RUL of the tested bearings for comparison, including SVM [9], DBN [24], multi-scale CNN (MCNN) [25] and convolutional LSTM (CLSTM) [26]. For the first three prognostics approaches, i.e., SVM, DBN and MCNN, the feature extraction and selection are first performed following [9,24,25]. Then, the selected features are input into the corresponding models for training and testing. Whereas for CLSTM, one-layer CNN is first used to extract the local robust features from vibration signals of bearings, and then these features are fed into bidirectional LSTM to perform the RUL estimation. Especially, to

Table 3

The effect of the dimensionality reduction ratio on the prediction performance of DSCN.

Reduction ratio r	Total model parameters	Training time (s)	Score (mean $\pm$ standard deviation)			RMSE (mean $\pm$ standard deviation)		
			Bearing 1_5	Bearing 2_5	Bearing 3_5	Bearing 1_5	Bearing 2_5	Bearing 3_5
4	59,617	2175.80	108.49 $\pm$ 16.11	150.96 $\pm$ 23.83	468.89 $\pm$ 55.27	6.88 $\pm$ 1.94	10.15 $\pm$ 2.42	11.83 $\pm$ 1.68
8	54,241	2082.24	102.98 $\pm$ 19.32	146.23 $\pm$ 25.04	497.61 $\pm$ 40.54	7.49 $\pm$ 1.06	9.06 $\pm$ 1.60	10.91 $\pm$ 1.71
16	51,553	2023.13	107.08 $\pm$ 14.94	134.80 $\pm$ 21.46	479.92 $\pm$ 49.64	6.67 $\pm$ 1.56	8.58 $\pm$ 1.69	10.41 $\pm$ 1.78
32	50,209	2010.35	108.50 $\pm$ 17.80	137.01 $\pm$ 28.80	475.48 $\pm$ 59.15	7.38 $\pm$ 1.05	9.62 $\pm$ 2.43	10.17 $\pm$ 1.88

The bold values mean that they are the best results among all results.

Table 4

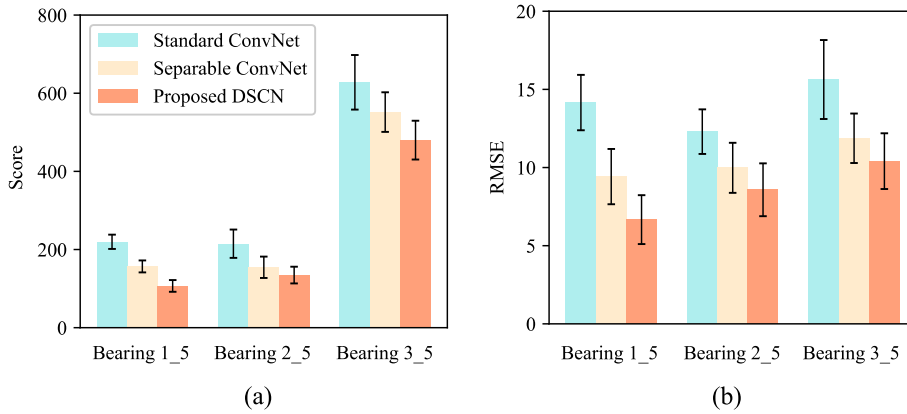
The effect of the network depth on the prediction performance of DSCN.

Number of blocks D	Total model parameters	Training time (s)	Score (mean $\pm$ standard deviation)			RMSE (mean $\pm$ standard deviation)		
			Bearing 1_5	Bearing 2_5	Bearing 3_5	Bearing 1_5	Bearing 2_5	Bearing 3_5
1	3137	995.89	435.87 $\pm$ 52.49	445.13 $\pm$ 49.25	1312.90 $\pm$ 134.46	16.46 $\pm$ 3.19	15.27 $\pm$ 4.32	20.38 $\pm$ 5.57
2	13,473	1239.21	162.91 $\pm$ 20.38	209.25 $\pm$ 19.69	573.02 $\pm$ 42.48	9.14 $\pm$ 1.87	9.93 $\pm$ 2.55	15.22 $\pm$ 2.02
3	51,553	2023.13	107.08 $\pm$ 14.94	134.80 $\pm$ 21.46	479.92 $\pm$ 49.64	6.67 $\pm$ 1.56	8.58 $\pm$ 1.69	10.41 $\pm$ 1.78
4	197,345	4403.73	101.13 $\pm$ 17.38	120.67 $\pm$ 21.98	419.15 $\pm$ 41.70	5.81 $\pm$ 1.26	8.10 $\pm$ 1.59	9.67 $\pm$ 1.25
5	767,457	6288.29	94.91 $\pm$ 17.05	121.03 $\pm$ 26.54	441.20 $\pm$ 36.10	6.39 $\pm$ 1.10	7.98 $\pm$ 1.06	10.19 $\pm$ 1.38

Table 5

Comparison of total model parameters and training time for three different prognostics networks.

Prognostics network	Total model parameters	Training time (s)
Standard ConvNet	272,171	4607.36
Separable ConvNet	48,865	1940.12
Proposed DSCN	51,553	2023.13

**Fig. 13.** Performance estimation results of three different prognostics networks. (a) Scoring function values. (b) RMSE values.

achieve more accurate RUL prediction results, hyperparameter tuning is also performed by 4-fold cross-validation for each prediction model.

Fig. 14 summarizes the performance estimation results of these five different prognostics approaches. The first observation is that for each of the three tested bearings, the deep learning models (i.e., DBN, MCNN, CLSTM and DSCN) consistently outperform the traditional machine model (i.e., SVM) in terms of both the score and RMSE. This is because the traditional machine learning models are built upon shallow learning architectures and thus they have limited ability to learn the degradation behavior of bearings. Contrarily, because of the outstanding representation learning ability, deep learning models are able to better mine valuable bearing degradation information and build more accurate RUL prediction models. Further, it can be observed from Fig. 14 that the proposed DSCN gets the lowest score values and the smallest RMSE values among all of the deep prognostics models, which indicates that DSCN provides the most accurate RUL estimation results for each bearing. Consequently, the proposed DSCN is superior to the other four prognostics approaches in RUL prediction of bearings. This performance improvement verifies again the benefits from the separable convolutions and the feature response recalibrations.

4.2. Case study 2: Public C-MAPSS datasets

4.2.1. C-MAPSS datasets

To further validate the effectiveness and superiority of the proposed DSCN, the prognostics benchmark datasets, i.e., C-MAPSS datasets, are used in this case study. The C-MAPSS datasets are provided by the Prognostics Center of Excellence (PCoE) at NASA Ames Research Center and are available on the website of Prognostics Data Repository of PCoE [43]. As tabulated in Table 6, the C-MAPSS datasets are comprised of four sub-datasets that are simulated under different combinations of operational conditions and fault modes. The run-to-failure simulations of turbofan engines are conducted using C-MAPSS software, and the monitoring data from 21 sensors are acquired to reflect the failure progressions of turbofan engines. In

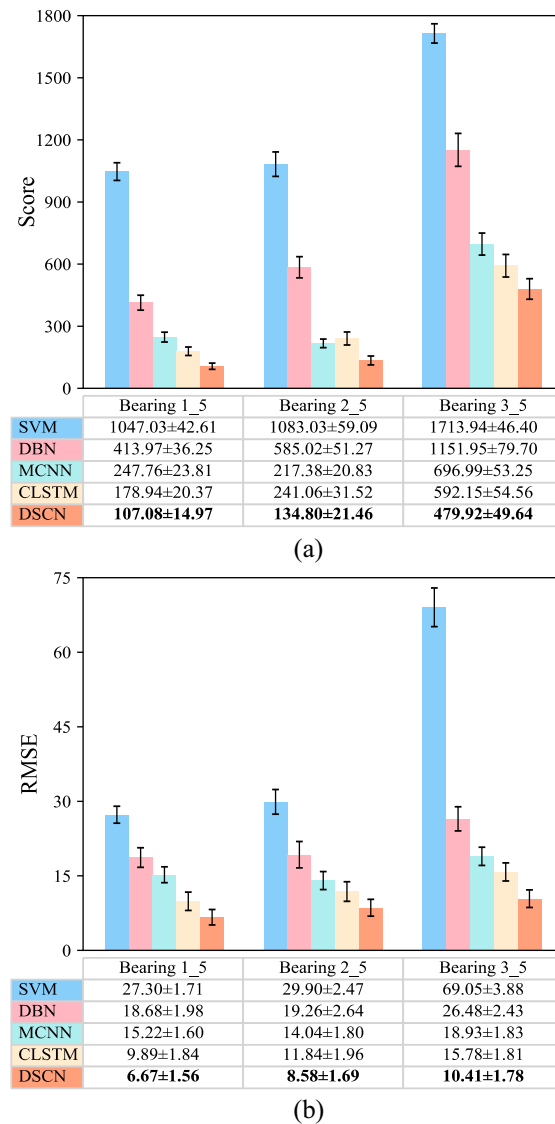


Fig. 14. Performance comparison of five different prognostics approaches on bearing datasets. (a) Scoring function values. (b) RMSE values.

particular, each sub-dataset is divided into one training dataset and one testing dataset. For training datasets, the complete run-to-failure data are provided, including the multi-sensor measurements and the corresponding ground truth RUL values. While for testing datasets, the monitoring data of each turbofan engine are truncated so as to predict the RUL.

4.2.2. Prognostics results, comparison and discussion

During the degradation simulations, a total of 21 variables are output from the C-MAPSS model. These variables represent the sensor responses from various components of the turbofan engine system. However, some of them have constant values

Table 6
Description of C-MAPSS datasets.

Dataset	Number of turbofan engines for training	Number of turbofan engines for testing	Number of operating conditions	Number of fault modes
FD001	100	100	1	1
FD002	260	259	6	1
FD003	100	100	1	2
FD004	249	248	6	2

Table 7

Performance comparison of the proposed DSCN and the six published studies on C-MAPSS datasets.

Dataset		RVM [38]	MODBNE [26]	LSTM1 [44]	LSTM2 [45]	BiLSTM [46]	CNN1 [38]	CNN2 [27]	Proposed DSCN
FD001	Score	1502.90	334.23	388.68	338	295	1286.7	273.70	260.67
	RMSE	23.80	15.04	16.74	16.14	13.65	18.45	12.61	10.95
FD002	Score	17,423	5585.34	10,654	4450	4130	13,570	10,412	4367.56
	RMSE	31.30	25.05	29.43	24.49	23.18	30.29	22.36	20.47
FD003	Score	1431.60	421.91	822.19	852	317	1596.2	284.1	246.55
	RMSE	22.37	12.51	18.07	16.18	13.74	19.82	12.64	10.62
FD004	Score	26,509	6557.62	6370.60	5550	5430	7886.4	12,466	5168.45
	RMSE	34.34	28.66	28.40	28.17	24.86	29.16	23.31	22.64

The bold values mean that they are the best results among all results.

during the whole operating life and thus cannot provide any useful degradation information for RUL prediction. Therefore, the sensor selection is firstly carried out following [44], where 14 sensors are selected as the inputs of prognostics models. Then, all sensor data are normalized using min-max scaling [28]. In addition, a time window of size 30 [28] and a piece-wise linear degradation assumption [45] are also employed in the data pre-processing. Correspondingly, the input volume size of DSCN is $30 \times 1 \times 14$.

In RUL prediction of turbofan engines, the proposed DSCN uses the same configuration tabulated in Table 2 except for the time window size. Meanwhile, the score function and RMSE are also utilized as performance evaluation metrics. Furthermore, to illustrate the superiority of the proposed approaches, the prediction results of DSCN are compared with the results reported in six published studies, including the results of relevance vector machine (RVM) and classic CNN (denoted as CNN1) from [40], the results of multi-objective deep belief network ensemble (MODBNE) from [28], the results of LSTM from [46] (denoted as LSTM1) and [47] (denoted as LSTM2), the results of bidirectional LSTM from [48] (denoted as BiLSTM), and the results of CNN without pooling layers (denoted as CNN2) from [29]. Table 7 shows the performance comparison results of the proposed DSCN and the six published studies on C-MAPSS datasets. It can be clearly seen that for each sub-dataset, the proposed DSCN achieves a lower score value and a smaller RMSE value compared with all other prognostics approaches, which signifies that the proposed DSCN performs much better in RUL prediction of turbofan engines. Furthermore, it is noticed that as compared to FD001 and FD003, the rest two sub-datasets have higher score values and larger RMSE values. This is because the sub-datasets of FD002 and FD004 are simulated under six different operating conditions, which increases the complexity of the prognostics problems. Particularly, for the two typical convolutional networks in [40] and [29], the increased complexity makes the representation learning more difficult, resulting in higher prediction score values than the other commonly used deep models, i.e., MODBNE, LSTM and BiLSTM. Whereas for the convolutional network proposed in this paper, because of the use of separable convolutions and the building of SE units, the competitive accuracy still can be obtained even though the complexity of the prognostics problems is increased. Based on the above two case studies, it is concluded that the proposed DSCN is able to accurately model the degradation processes of machines based on the raw multi-sensor data, and achieves better prediction performance in comparison with the existing prognostics approaches.

5. Conclusions

In this paper, a new deep prognostics network called DSCN is proposed for RUL prediction of machinery. The proposed DSCN directly uses the raw multi-sensor data as the inputs of the prognostics network. Then, a separable convolution operation is introduced to replace the standard convolution operation so as to effectively model the interrelationships of different sensor data. Meanwhile, in order to increase the sensitivity of the prognostics network to informative feature maps, an SE unit is constructed behind the separable convolutional layer to perform adaptive feature response recalibrations. After that, a separable convolutional building block is built based on the separable convolutions and the SE unit. In particular, residual connections and pre-activations are also employed in the separable convolutional building block to ease the training of the network and relieve the problem of overfitting. Through stacking multiple separable convolutional building blocks, the high-level representations are learned from the input data and the RUL is finally estimated by feeding the learned representations to the last fully-connected layer. The proposed DSCN is experimentally validated using the vibration data from accelerated degradation tests of rolling element bearings and the public C-MAPSS datasets, and is also compared with some state-of-the-art prognostics approaches. The experimental results show that the proposed DSCN achieves high RUL prediction accuracy and is superior to traditional data-driven approaches and typical deep learning models.

The advantages of the proposed DSCN in practical applications are summarized as follows. 1) The proposed DSCN directly uses raw multi-sensor data as inputs, getting rid of the complex process of manual feature extraction and selection. In fact, many existing deep prognostics networks still need to manually extract and select some discriminative features from raw sensor data. Such procedures, however, are time-consuming and labor-intensive while requiring explicit prior knowledge and expertise. Moreover, features used in prognostic are usually designed case by case and so their generalization ability may be weak. Therefore, for practical applications, it is very important and meaningful to directly use raw sensor data as the inputs of a deep prognostics network. 2) The proposed DSCN explicitly considers the correlations of different sensor data.

In industrial applications, a machine is usually monitored by multiple sensors. Accordingly, the acquired multi-sensor data have two distinct characteristics. One is that different sensor data contain different degrees of degradation information, and some of these sensor data may be sensitive to machine degradation, but others may be not. The other one is that these sensor data are able to capture the propagation and interaction of faults among different components. In the proposed DSCN, these corrections are sufficiently considered during network construction. Separable convolutions are first introduced to model the interrelationships of different sensor data, and then SE units are constructed to emphasize the important degradation information. Thus, the proposed DSCN is able to effectively improve the prediction accuracy in practical applications.

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References

- [1] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, J. Lin, Machinery health prognostics: a systematic review from data acquisition to RUL prediction, *Mech. Syst. Sig. Process.* 104 (2018) 799–834.
- [2] K. Javed, R. Gouriveau, N. Zerhouni, State of the art and taxonomy of prognostics approaches, trends of prognostics applications and open issues towards maturity at different technology readiness levels, *Mech. Syst. Sig. Process.* 94 (2017) 214–236.
- [3] L. Ren, X. Cheng, X. Wang, J. Cui, L. Zhang, Multi-scale dense gate recurrent unit networks for bearing remaining useful life prediction, *Future Generation Comput. Syst.* 94 (2019) 601–609.
- [4] R. Zhao, R. Yan, Z. Chen, K. Mao, P. Wang, R.X. Gao, Deep learning and its applications to machine health monitoring, *Mech. Syst. Sig. Process.* 115 (2019) 213–237.
- [5] X. Si, Z. Zhang, C. Hu, *Data-driven Remaining Useful Life Prognosis Techniques*, Springer, 2017.
- [6] J. Cui, L. Ren, X. Wang, L. Zhang, Pairwise comparison learning based bearing health quantitative modeling and its application in service life prediction, *Future Generation Comput. Syst.* 97 (2019) 578–586.
- [7] D. Wang, K.-L. Tsui, Theoretical investigation of the upper and lower bounds of a generalized dimensionless bearing health indicator, *Mech. Syst. Sig. Process.* 98 (2018) 890–901.
- [8] S.A. Aye, P.S. Heyns, An integrated Gaussian process regression for prediction of remaining useful life of slow speed bearings based on acoustic emission, *Mech. Syst. Sig. Process.* 84 (2017) 485–498.
- [9] T.H. Loutas, D. Roulas, G. Georgoulas, Remaining useful life estimation in rolling bearings utilizing data-driven probabilistic E-support vectors regression, *IEEE Trans. Reliab.* 62 (2013) 821–832.
- [10] J.B. Ali, B. Chebel-Morello, L. Saidi, S. Malinowski, F. Fnaiech, Accurate bearing remaining useful life prediction based on Weibull distribution and artificial neural network, *Mech. Syst. Sig. Process.* 56–57 (2015) 150–172.
- [11] K. Javed, R. Gouriveau, N. Zerhouni, P. Nectoux, Enabling health monitoring approach based on vibration data for accurate prognostics, *IEEE Trans. Ind. Electron.* 62 (2015) 647–656.
- [12] P. Lim, C.K. Goh, K.C. Tan, A novel time series-histogram of features (TS-HoF) method for prognostic applications, *IEEE Trans. Emerging Topics Comput. Intel.* 2 (2018) 204–213.
- [13] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, *Nature* 521 (2015) 436–444.
- [14] G.E. Hinton, S. Osindero, Y.-W. Teh, A fast learning algorithm for deep belief nets, *Neural Comput.* 18 (2006) 1527–1554.
- [15] Y. LeCun, L. Bottou, Y. Bengio, P. Haffner, Gradient-based learning applied to document recognition, *Proc. IEEE* 86 (1998) 2278–2324.
- [16] Y. Bengio, P. Simard, P. Frasconi, Learning long-term dependencies with gradient descent is difficult, *IEEE Trans. Neural Networks* 5 (1994) 157–166.
- [17] M. Bojarski, D.D. Testa, D. Dworakowski, B. Firner, B. Flepp, P. Goyal, L.D. Jackel, M. Monfort, U. Muller, J. Zhang, X. Zhang, J. Zhao, K. Zieba, End to end learning for self-driving cars, 2016, arXiv preprint arXiv:1604.07316.
- [18] T. Afouras, J.S. Chung, A. Senior, O. Vinyals, A. Zisserman, Deep audio-visual speech recognition, *IEEE Trans. Pattern Anal. Mach. Intell.* (2018) 1–11.
- [19] M. Gardner, J. Grus, M. Neumann, O. Tafjord, P. Dasigi, N. Liu, M. Peters, M. Schmitz, L. Zettlemoyer, AllenNLP: A deep semantic natural language processing platform, 2018, arXiv preprint arXiv:1803.07640.
- [20] H. Chen, O. Engkvist, Y. Wang, M. Olivecrona, T. Blaschke, The rise of deep learning in drug discovery, *Drug Discov. Today* 23 (2018) 1241–1250.
- [21] S. Khan, T. Yairi, A review on the application of deep learning in system health management, *Mech. Syst. Sig. Process.* 107 (2018) 241–265.
- [22] L. Guo, Y. Lei, S. Xing, T. Yan, N. Li, Deep convolutional transfer learning network: a new method for intelligent fault diagnosis of machines with unlabeled data, *IEEE Trans. Ind. Electron.* 66 (2018) 7316–7325.
- [23] L. Ren, Y. Sun, J. Cui, L. Zhang, Bearing remaining useful life prediction based on deep autoencoder and deep neural networks, *J. Manuf. Syst.* 48 (2018) 71–77.
- [24] J. Deutsch, D. He, Using deep learning-based approach to predict remaining useful life of rotating components, *IEEE Trans. Syst. Man Cyber. Syst.* 48 (2017) 11–20.
- [25] J. Zhu, N. Chen, W. Peng, Estimation of bearing remaining useful life based on multiscale convolutional neural network, *IEEE Trans. Ind. Electron.* 66 (2018) 3208–3216.
- [26] R. Zhao, R. Yan, J. Wang, K. Mao, Learning to monitor machine health with convolutional bi-directional LSTM networks, *Sensors* 17 (2017) 273.
- [27] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: 2016 IEEE Conference on Computer Vision and Pattern Recognition, IEEE, 2016, pp. 1–9.
- [28] C. Zhang, P. Lim, A.-K. Qin, K.C. Tan, Multiobjective deep belief networks ensemble for remaining useful life estimation in prognostics, *IEEE Trans. Neural Networks Learn. Syst.* 28 (2017) 1–13.
- [29] X. Li, Q. Ding, J.-Q. Sun, Remaining useful life estimation in prognostics using deep convolution neural networks, *Reliab. Eng. Syst. Saf.* 172 (2018) 1–11.
- [30] Y. Wu, M. Yuan, S. Dong, L. Lin, Y. Liu, Remaining useful life estimation of engineered systems using vanilla LSTM neural networks, *Neurocomputing* 275 (2017) 167–179.
- [31] F. Mamalet, C. Garcia, Simplifying ConvNets for fast learning, in: The 22nd International Conference on Artificial Neural Networks, Springer, 2012, pp. 58–65.
- [32] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks, 2017, arXiv preprint arXiv: 1709.01507.
- [33] K. He, X. Zhang, S. Ren, J. Sun, Identity mappings in deep residual Networks, in: The 14th European Conference on Computer Vision, Springer, 2016, pp. 630–645.
- [34] S. Ioffe, C. Szegedy, Batch normalization: Accelerating deep network training by reducing internal covariate shift, 2015, arXiv preprint arXiv:1502.03167.
- [35] V. Nair, G.E. Hinton, Rectified linear units improve restricted Boltzmann machines, in: Proceedings of the 27th International Conference on Machine Learning, Omnipress, 2010, pp. 807–814.

- [36] B. Wang, Y. Lei, N. Li, N. Li, A hybrid prognostics approach for estimating remaining useful life of rolling element bearings, *IEEE Trans. Reliab.* (2018) 1–12.
- [37] A. Saxena, K. Goebel, D. Simon, N. Eklund, Damage propagation modeling for aircraft engine run-to-failure simulation, in: 2008 International Conference on Prognostics and Health Management, IEEE, 2008, pp. 1–9.
- [38] Y. Gu, B. Wylie, S. Boyte, J. Picotte, D. Howard, K. Smith, K. Nelson, An optimal sample data usage strategy to minimize overfitting and underfitting effects in regression tree models based on remotely-sensed data, *Rem. Sens.* 8 (2016) 943.
- [39] X. Si, W. Wang, C.-H. Hu, D.-H. Zhou, Remaining useful life estimation – a review on the statistical data driven approaches, *Eur. J. Oper. Res.* 213 (2011) 1–14.
- [40] G.S. Babu, P. Zhao, X. Li, Deep convolutional neural network based regression approach for estimation of remaining useful life, in: 21st International Conference on Database Systems for Advanced Applications, Springer, 2016, pp. 214–228.
- [41] D.P. Kingma, J.L. Ba, Adam: A method for stochastic optimization, 2015, arXiv preprint arXiv:1412.6980.
- [42] R.K. Singleton, E.G. Strangas, S. Aviyente, Extended Kalman filtering for remaining-useful-life estimation of bearings, *IEEE Trans. Ind. Electron.* 62 (2015) 1781–1790.
- [43] A. Saxena, K. Goebel, Turbofan engine degradation simulation data set, NASA Ames Prognostics Data Repository (2008). <https://ti.arc.nasa.gov/tech/dash/groups/pcoe/prognostic-data-repository/#turbofan>.
- [44] P. Lim, C.K. Goh, K.C. Tan, P. Dutta, Estimation of remaining useful life based on switching Kalman filter neural network ensemble, in: The 2014 Annual Conference of the Prognostics and Health Management Society, Prognostics and Health Management Society, 2014, pp. 2–9.
- [45] F.O. Heimes, Recurrent neural networks for remaining useful life estimation, in: 2008 International Conference on Prognostics and Health Management, IEEE, 2008, pp. 1–6.
- [46] C.-S. Hsu, J.-R. Jiang, Remaining useful life estimation using long short-term memory deep learning, in: 2018 IEEE International Conference on Applied System Invention, IEEE, 2018, pp. 1–4.
- [47] S. Zheng, K. Ristovski, A. Farahat, C. Gupta, Long short-term memory network for remaining useful life estimation, in: 2017 IEEE International Conference on Prognostics and Health Management, IEEE, 2017, pp. 1–8.
- [48] J. Wang, G. Wen, S. Yang, Y. Liu, Remaining useful life estimation in prognostics using deep bidirectional LSTM neural network, in: 2018 Prognostics and System Health Management Conference (PHM-Chongqing), IEEE, 2018, pp. 1037–1042.